

F - Critical Misread

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Time Limit: 2 sec / Memory Limit: 1024 MiB

Score : 525 points



Problem Statement

You are given K strings S_i consisting of lowercase English letters.

Find the number, modulo 998244353, of strings of length N consisting of lowercase English letters that do not contain any of $S_1, S_2, \dots, S_K$ as a substrin (contiguous subsequence).



Constraints

- N is an integer between 1 and 10^9 , inclusive.
- K is an integer between 1 and 10, inclusive.
- S_i is a string consisting of lowercase English letters with length between 1 and 10, inclusive.



Input

The input is given from Standard Input in the following format:

```
N K
S1
S2
⋮
SK
```



Output

Output the answer.



Sample Input 1

Copy

```
3 2
aa
ab
```

Cop

Sample Output 1

Copy

17474

Cop

- The number of strings of length 3 consisting of lowercase English letters is $26^3 = 17576$.
- The number of strings of length 3 consisting of lowercase English letters whose first two characters are aa is 26.
- The number of strings of length 3 consisting of lowercase English letters whose last two characters are aa is 26.
- The number of strings of length 3 consisting of lowercase English letters whose first two characters are ab is 26.
- The number of strings of length 3 consisting of lowercase English letters whose last two characters are ab is 26.
- The number of strings of length 3 consisting of lowercase English letters whose entirety is aaa is 1.
- The number of strings of length 3 consisting of lowercase English letters whose entirety is aab is 1.
- Combining the above facts, it can be confirmed that the answer for this test case is 17474.

Sample Input 2

Copy

1 1
ab

Cop

Sample Output 2

Copy

26

Cop

Sample Input 3

Copy

1000000000 10
contest
tester
error
orange
angel
elegant
antitese
sextuple
pleasure
surely

Cop

Sample Output 3

Copy

698570468

Cop

Find the count modulo 998244353.

1

► Custom test data (auto-saving)



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Remaining Time
01:23:41